

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS:

Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
B. `<video src="video.mp4" controls></video>`

- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

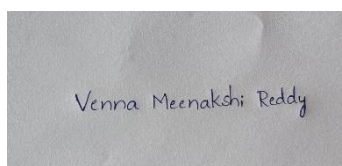
1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

/ VENNA MEENAKSHI REDDY (192471048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Venna Meenakshi Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANSWER:

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B.

```
<video src="video.mp4" controls></video>
```

Justification:

The <video> element is the standard HTML5 tag used to embed video content into a web page. It supports multiple video formats such as MP4, WebM, and Ogg and provides built-in playback controls like Play, Pause, Volume, Full Screen, and Seek when the controls attribute is included.

It is supported by all modern web browsers and improves user experience by allowing users to control video playback without additional plugins.

Why the other options are incorrect:

- **A. <media>** – Not a valid HTML element.
- **C. <movie>** – Not part of the HTML specification.
- **D. <vid>** – Invalid HTML tag.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

Justification:

The POST method is the most appropriate method for submitting registration forms because it sends data within the HTTP request body instead of the URL. This protects sensitive information such as usernames, passwords, email addresses, and personal details from being exposed in browser history or URLs.

POST also supports sending large amounts of data and is commonly used when creating new resources or storing user information in a database.

Why the other options are incorrect:

- **A. GET** – Sends data through the URL, making sensitive information visible.
- **C. PUT** – Primarily used to update existing resources in REST APIs.
- **D. DELETE** – Used to remove resources from the server.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. <nav>

Justification:

The <nav> element is the semantic HTML5 element specifically designed to contain navigation links such as menus, sidebars, breadcrumbs, or page indexes.

Using <nav> improves:

- Accessibility for screen readers.
- Search Engine Optimization (SEO).
- Code readability and maintainability.
- Compliance with modern HTML5 standards.

Why the other options are incorrect:

- **A. <div>** – Generic container with no semantic meaning.
- **C. <section>** – Used to group related content, not specifically navigation.
- **D. <header>** – Represents introductory content and may contain navigation, but it is not dedicated solely to navigation.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: B.

```
<note>  
  <to>User</to>  
  <from>Admin</from>  
</note>
```

Justification:

This XML document follows all XML syntax rules:

- Every opening tag has a matching closing tag.
- Elements are properly nested.
- There is a single root element (<note>).
- Tags are correctly closed.

Errors in the other options:

Option A

```
<note><to>User</to><from>Admin</from>
```

Error: Missing the closing </note> tag.

Option C

```
<note><to>User</to><from>Admin</from></note>
```

with

```
<note><to>User</from></note>
```

Error: Mismatched opening and closing tags (<to> is incorrectly closed with </from>).

Option D

```
<note to="User"></note>
```

Errors:

- Missing the closing angle bracket (>).
- Invalid XML syntax due to an incomplete closing tag.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP (HyperText Transfer Protocol)

Justification:

HTTP is the standard **application-layer protocol** used for communication between web browsers (clients) and web servers. It enables the transfer of web pages, images, videos, CSS files, JavaScript files, and other web resources.

For secure communication, websites commonly use **HTTPS**, which is HTTP combined with SSL/TLS encryption to ensure confidentiality, integrity, and authentication.

Why the other options are incorrect:

- **A. FTP (File Transfer Protocol)** – Used for transferring files between computers.
- **C. SMTP (Simple Mail Transfer Protocol)** – Used for sending emails.
- **D. SNMP (Simple Network Management Protocol)** – Used for monitoring and managing network devices.